Skills Review

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Example No. 15: This program prompts the user to input the required information. To facilitate clean output, functions 06h will be used to clear the screen, 02h to set the screen location to display the prompts, 09h to display, and 0Ah to request input from the user. Save the program using sample15.asm as the filename

Program

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.data

wn      DB 'What is your name?',20h,'$'
ag      DB 'how old are you?',20h,'$'
bd      DB 'When is  your birthday?',20h,'$'
gd      DB 'What is your gender?',20h,'$'
rpt     DB 'Repeat? Press N to Exit. ',20h,20h,'$'

wnpar   label byte              ; parameter list for name
wnlen   DB 20                   ; maximum number of characters
actwnlen DB ?                   ; actual length entered
wnfld   DB 20 dup(' ')          ; characters entered

agpar   label byte              ; parameter list for age
aglen   DB 10
actaglen DB ?
agfld   DB 10 dup(' ')

bdpar   label byte              ; parameter list for birthday
bdlen   DB 20
actbdlen DB ?
bdfld   DB 20 dup(' ')

gdpar   label byte              ; parameter list for gender
gdlen   DB 10
actgdlen DB ?
gdfld   DB 10 dup(' ')
```

```
.code
start:
        mov ax, @data           ; initialize the data segment
        mov ds, ax

ulit:
        ;** clear the screen **
        mov ah, 06h             ; Function 06h - scroll up / clear window
        mov al, 00h             ; AL = 0 - clear the whole window
        mov bh, 07h             ; attribute - white on black
        mov cx, 0000h           ; upper left corner (row 0, col 0)
        mov dx, 184Fh           ; lower right corner (row 24, col 79)
        int 10h

        ;** first prompt **
        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00h             ; page 0
        mov dx, 0820h           ; row 8, col 32
        int 10h

        mov ah, 09h             ; Function 09h - display string
        mov dx, offset wn       ; "What is your name?"
        int 21h

        ;** response to first prompt **
        mov ah, 0Ah             ; Function 0Ah - buffered string input
        mov dx, offset wnpar
        int 21h

        ;** second prompt **
        mov ah, 02h
        mov bh, 00h
        mov dx, 0920h           ; row 9, col 32
        int 10h

        mov ah, 09h
        mov dx, offset ag       ; "how old are you?"
        int 21h
```

```
        ;** response to second prompt **
        mov ah, 0Ah
        mov dx, offset agpar
        int 21h

        ;** third prompt **
        mov ah, 02h
        mov bh, 00h
        mov dx, 0A20h          ; row 10, col 32
        int 10h

        mov ah, 09h
        mov dx, offset bd      ; "When is your birthday?"
        int 21h

        ;** response to third prompt **
        mov ah, 0Ah
        mov dx, offset bdpar
        int 21h

        ;** fourth prompt **
        mov ah, 02h
        mov bh, 00h
        mov dx, 0B20h          ; row 11, col 32
        int 10h

        mov ah, 09h
        mov dx, offset gd      ; "What is your gender?"
        int 21h

        ;** response to fourth prompt **
        mov ah, 0Ah
        mov dx, offset gdpar
        int 21h

        ;** repeat prompt **
        mov ah, 02h
        mov bh, 00h
        mov dx, 0E24h          ; row 14, col 36
        int 10h
```

```
        mov ah, 09h
        mov dx, offset rpt      ; "Repeat? Press N to Exit."
        int 21h

        mov ah, 01h             ; Function 01h - read a key with echo
        int 21h

        cmp al, 'N'             ; is it N?
        je tapos                ; if so, end the program
        cmp al, 'n'             ; also accept lowercase n
        je tapos
        jmp ulit                ; any other key - repeat

tapos:  mov ah, 4Ch             ; terminate program
        int 21h

end start
```

Output

